UNIVERSITY OF CENTRAL PUNJAB

Faculty of Information Technology

Cyber Security

Week 5 — Malware Threats in Depth

What Malware Is • Types of Malware • Impact on Systems
Stages of a Malware Attack • Payloads • Analysis Tools & Practice

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QUICK RECAP

Where We Are So Far

Week 1–2	Cybersecurity basics, the CIA Triad, threats, and attack vectors.
Week 3	Vulnerabilities, the OWASP Top 10, and CVE / CVSS / NVD.
Week 4	Common threats, DoS/DDoS service attacks, and reconnaissance.
This week	We go deep on MALWARE — the harmful software behind so many attacks.

WEEK 5 ROADMAP

What We Will Cover This Week

1

Malware & Its Types

What malware is and the main families you must know

2

Impact of Malware

The damage malware does to systems, data and people

3

Stages of a Malware Attack

How malware spreads, hides, and delivers its payload

4

Defense, Tools & Practice

Protection, analysis tools, and safe hands-on labs

DEFINITION

What is Malware?

Malware ("malicious software") is any program written to damage, disrupt, spy on, or take control of a computer without the owner's consent.

In simple words: malware is software with bad intentions. It often hides inside something that looks safe — a file, an app, or a link.

Damage

Deletes, corrupts or encrypts your files and programs

Steal

Secretly copies passwords, data, and keystrokes

Control

Takes over your device to obey the attacker

INFECTION

How Does Malware Spread?

Email attachments

Opening an infected file in a phishing email.

Malicious downloads

"Free" or cracked software carrying hidden malware.

Infected websites

Drive-by downloads that install malware automatically.

USB / removable media

Plugging in an infected USB stick.

Fake apps

Malicious mobile apps disguised as real ones.

Network worms

Malware that spreads computer-to-computer by itself.

THE MAIN FAMILIES

Types of Malware — Overview

We'll define each of these in the coming slides:

Virus	Worm	Trojan	Ransomware	Spyware
Adware	Rootkit	Botnet	Keylogger	Fileless

Some malware fits more than one type — for example, a worm can also carry ransomware.

Types of Malware (1 of 3)

Virus	**Attaches to a file and spreads when that file is opened/run.** **Example:** An infected document that infects others once opened.
Worm	**Self-replicating — spreads across networks with no human help.** **Example:** Spreads PC-to-PC on its own, clogging the whole network.
Trojan	**Looks like a useful program but hides harm inside.** **Example:** A "free PDF reader" that secretly opens a backdoor.
Ransomware	**Encrypts your files and demands payment to unlock them.** **Example:** WannaCry locked files worldwide and demanded Bitcoin.

Types of Malware (2 of 3)

Spyware	**Secretly watches and reports what you do.** **Example:** Software that records the websites and data you enter.
Keylogger	**Records every keystroke to steal passwords & messages.** **Example:** Captures your bank login exactly as you type it.
Adware	**Floods you with unwanted ads and pop-ups.** **Example:** Endless pop-ups after installing a free program.
Rootkit	**Hides deep in the system to avoid detection.** **Example:** Gives an attacker hidden admin control of your PC.

Types of Malware (3 of 3)

Botnet

Many infected devices controlled together by an attacker.

Example: Thousands of "zombie" PCs used to launch a DDoS attack.

Fileless malware

Runs in memory using trusted tools — leaves few traces.

Example: Abusing PowerShell so nothing is written to disk.

Scareware

Fake warnings that trick you into paying or installing.

Example: "Your PC has 5 viruses! Buy this cleaner now!"

CLEAR THE CONFUSION

Virus vs Worm vs Trojan

VIRUS

Nature
Needs a host file

Spreading
Needs YOU to open/run it

Key point
Spreads by attaching to files

WORM

Nature
Stands alone

Spreading
Spreads by itself

Key point
Copies across networks automatically

TROJAN

Nature
Disguised as useful

Spreading
Needs YOU to install it

Key point
Doesn't self-spread — it tricks you

FOCUS

Ransomware — A Closer Look

Ransomware encrypts a victim's files, then demands a ransom (often in cryptocurrency) for the key to unlock them.

Step 1	Infects the system (often via a phishing email or bad download).
Step 2	Quietly encrypts documents, photos and databases.
Step 3	Shows a ransom note demanding payment by a deadline.
Step 4	May threaten to leak the data too ("double extortion").

REAL-WORLD CASES

Famous Malware in History

ILOVEYOU (2000)

A worm spread by email that hit millions in hours.

Example: An email "love letter" attachment that overwrote files.

Stuxnet (2010)

A nation-state worm that sabotaged industrial systems.

Example: Damaged centrifuges by targeting their controllers.

WannaCry (2017)

Ransomware that spread worldwide using a Windows flaw.

Example: Locked hospital and business systems for ransom.

Mirai (2016)

Malware that built a botnet from IoT devices.

Example: Hijacked cameras & routers to launch record DDoS attacks.

PART 2

The Impact of Malware

What actually happens to systems, data, and people when malware hits.

IMPACT

How Malware Affects a System

Slows performance

The device becomes slow, freezes, or crashes often.

Data loss/damage

Files get deleted, corrupted, or encrypted.

Stolen information

Passwords, banking and personal data are copied.

Unauthorized control

Attackers use your device remotely as a "bot."

Spreads further

Infects other devices on the same network.

Financial loss

Ransom payments, fraud, and recovery costs.

IMPACT

Wider Impact on People & Organizations

Privacy loss

Private messages, photos and identities exposed.

Business downtime

Companies stop operating while systems are down.

Reputation damage

Customers lose trust after a public infection.

Legal penalties

Fines for failing to protect user data.

Emotional stress

Victims feel violated, anxious, and helpless.

Critical services hit

Hospitals, power and transport can be disrupted.

DETECTION

Signs Your Device May Be Infected

Sudden slowness

The device lags or freezes for no clear reason.

Pop-ups & ads

Unexpected ads or fake warnings appear.

Programs crash

Apps close, or new unknown programs appear.

Files changed

Files are missing, renamed, or won't open.

High activity

Disk, network or CPU busy while you're idle.

Disabled security

Antivirus or updates get turned off by themselves.

PART 3

Stages of a Malware Attack

The life of a malware infection, from delivery to damage.

THE LIFECYCLE

Stages of a Malware Attack



Key idea

Every stage is a chance to stop the attack — block delivery, prevent execution, detect installation, or limit the spread.

STAGES EXPLAINED

Each Stage in Detail

Delivery

How malware reaches the target.

Example: A phishing email attachment or an infected USB.

Execution

The malicious code starts running.

Example: The victim double-clicks the file, launching it.

Installation

Malware embeds itself to survive restarts.

Example: It adds itself to startup so it runs every boot.

Spreading

It moves to other files, users or devices.

Example: A worm scans the network for new victims.

THE DAMAGE PART

What is a Payload?

The payload is the part of malware that actually carries out the harmful action — the "what it does" once it is inside.

Analogy: if malware is a delivery truck, the payload is the cargo it drops off — the actual damage.

Encrypt files

Lock data and demand a ransom (ransomware payload).

Steal data

Copy passwords and files back to the attacker.

Create a backdoor

Leave a secret way for the attacker to return.

Destroy / wipe

Delete files or make the system unusable.

PART 4

Defending Against Malware

How to prevent, detect, and respond to malware — plus tools and safe practice.

DEFENSE · PREVENT

How to Prevent Malware

Antivirus / EDR

Detect and block known malware automatically.

Update software

Patches close the holes malware exploits.

Think before clicking

Don't open unknown links, files, or USBs.

Download from trusted sources

Avoid cracked or pirated software.

Use a firewall

Block malicious network connections.

Least privilege

Limit what malware can do if it does get in.

If a Device Gets Infected

Step 1	Disconnect from the network to stop it spreading.
Step 2	Don't pay ransoms — it funds crime and may not work.
Step 3	Run a full antivirus / anti-malware scan.
Step 4	Remove the malware or wipe and reinstall the system.
Step 5	Restore files from a clean backup.
Step 6	Change passwords and investigate how it got in.

BEST PRACTICE

Backups — Your Safety Net

Good backups are the single best defense against ransomware — if your files are safely copied, you can restore instead of paying.

The 3-2-1 Backup Rule

3 copies

Keep at least three copies of important data.

2 media types

Store them on two different types of storage.

1 offsite

Keep one copy off-site or offline (safe from ransomware).

TOOLS · ANALYSIS

Malware Analysis & Protection Tools

Tools that professionals use to detect, study, and stop malware. (For learning — handle real malware only in a safe lab.)

VirusTotal

Upload a file or URL to scan it with 70+ antivirus engines.

Windows Defender / EDR

Built-in and enterprise tools that block malware in real time.

Any.Run / Cuckoo Sandbox

Run suspicious files safely and watch what they do.

Process Explorer / Autoruns

See running processes and what starts at boot (Sysinternals).

Wireshark

Spot malware "calling home" by inspecting network traffic.

Malwarebytes

Popular tool to detect and remove malware.

TOOLS · PRACTICE

Safe Ways to Study Malware

NEVER run real malware on your normal computer. Use isolated, disposable environments only.

Isolated VM (VirtualBox)

A disposable virtual machine with no network access.

FlareVM / REMnux

Ready-made analysis environments for studying malware.

TryHackMe / Hack The Box

Guided, legal rooms on malware analysis basics.

theZoo / MalwareBazaar

Malware samples for research — for advanced, careful use only.

EICAR test file

A harmless "fake virus" to safely test your antivirus.

VirusTotal (online)

Analyze suspicious files without running them yourself.

PRACTICE ACTIVITY 1

Test Your Antivirus Safely (EICAR)

A 100% safe activity — the EICAR file is NOT a real virus; it only triggers antivirus alerts for testing.

Step 1	Read what the EICAR test file is (an industry-standard safe test).
Step 2	Check that your antivirus is turned on.
Step 3	Create or download the EICAR test string into a text file.
Step 4	Watch your antivirus detect and quarantine it instantly.
Step 5	Discuss: how did the antivirus recognize it so fast?

PRACTICE ACTIVITY 2

Analyze a File with VirusTotal

Learn to check suspicious files WITHOUT running them — a real skill used by professionals every day.

Step 1	Go to virustotal.com in your browser.
Step 2	Upload a safe sample file (or paste a URL you want to check).
Step 3	See how many antivirus engines flag it, and how.
Step 4	Explore the 'Details' and 'Behavior' tabs to see what it does.
Step 5	Learn to read the verdict before opening any unknown file.

QUICK REFERENCE

Key Terms at a Glance

Malware

Software built to cause harm.

Virus

Malware that attaches to a host file.

Worm

Self-spreading malware, no host needed.

Trojan

Malware disguised as a useful program.

Ransomware

Encrypts files and demands payment.

Payload

The part that does the actual damage.

Botnet

A network of infected "zombie" devices.

Sandbox

A safe, isolated place to run suspicious files.

RECAP

Week 5 Summary

- Malware is software built to damage, steal, spy on, or control a device without consent.
- It spreads via email attachments, downloads, infected sites, USBs, fake apps and network worms.
- Main types: virus, worm, trojan, ransomware, spyware, keylogger, adware, rootkit, botnet, fileless.
- Impact ranges from slow devices and data loss to stolen data, downtime, and serious financial loss.
- A malware attack has stages: delivery, execution, installation, payload, and spread — each is a chance to stop it.
- Defend with antivirus, updates, careful clicking, least privilege and good backups (the 3-2-1 rule).
- Study malware ONLY in isolated labs; use VirusTotal and the safe EICAR test file to practise.

CHECK YOUR UNDERSTANDING

Review Questions

1. Define malware. Name five different types and describe each briefly.
2. Explain the difference between a virus, a worm, and a trojan.
3. What is ransomware, and why are backups the best defense against it?
4. List four ways malware can affect a system or an organization.
5. Describe the stages of a malware attack. What is a payload?
6. Name two safe, legal ways to study or test malware, and why they are safe.

WEEK 6 PREVIEW

Coming Next: Network Defenses

- Network defenses and firewalls
- Intrusion Detection & Prevention Systems (IDS/IPS)
- Benefits, limitations, and best practices